



USE CASE - #3

Retail: Sales & Customer Analytics

Presented By:

Team Snow Admiral

Team Members:

Kawinkumar B

Ganesh B





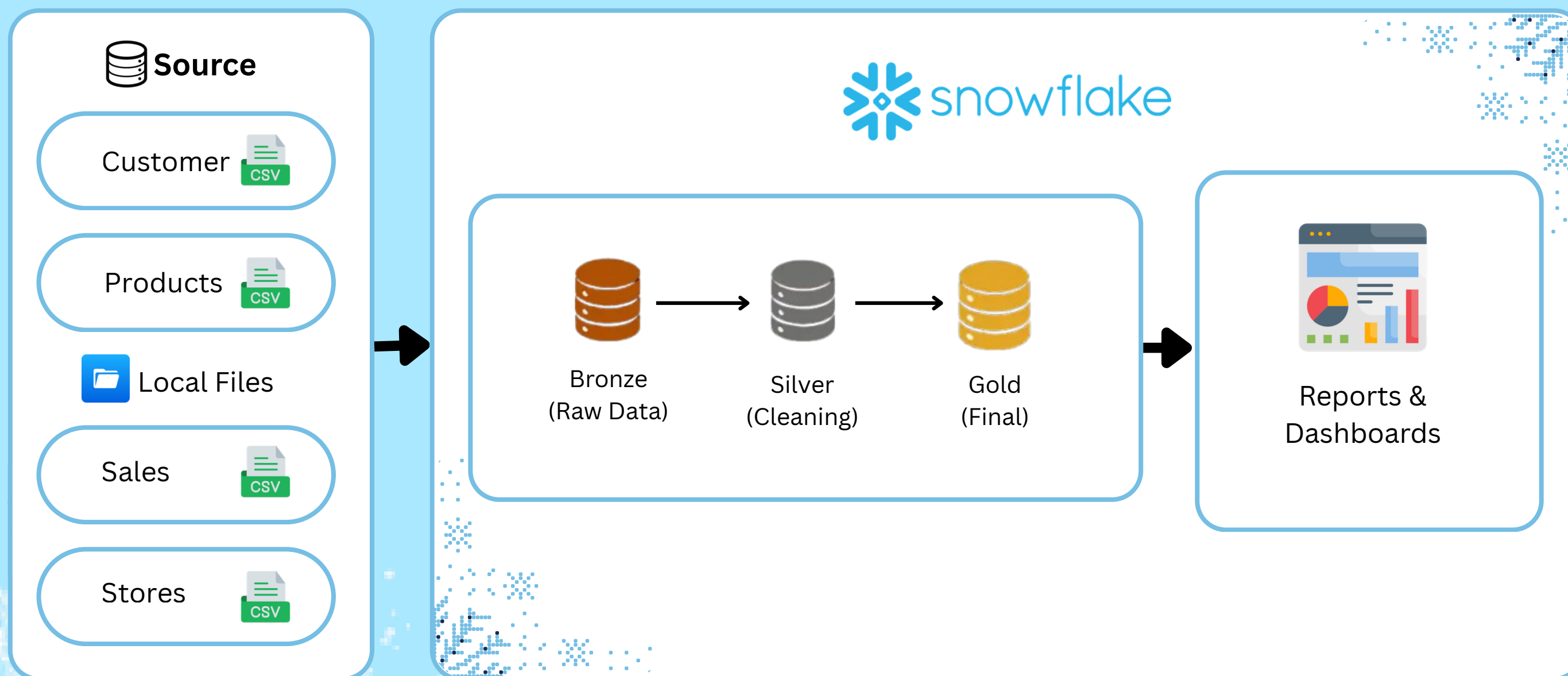
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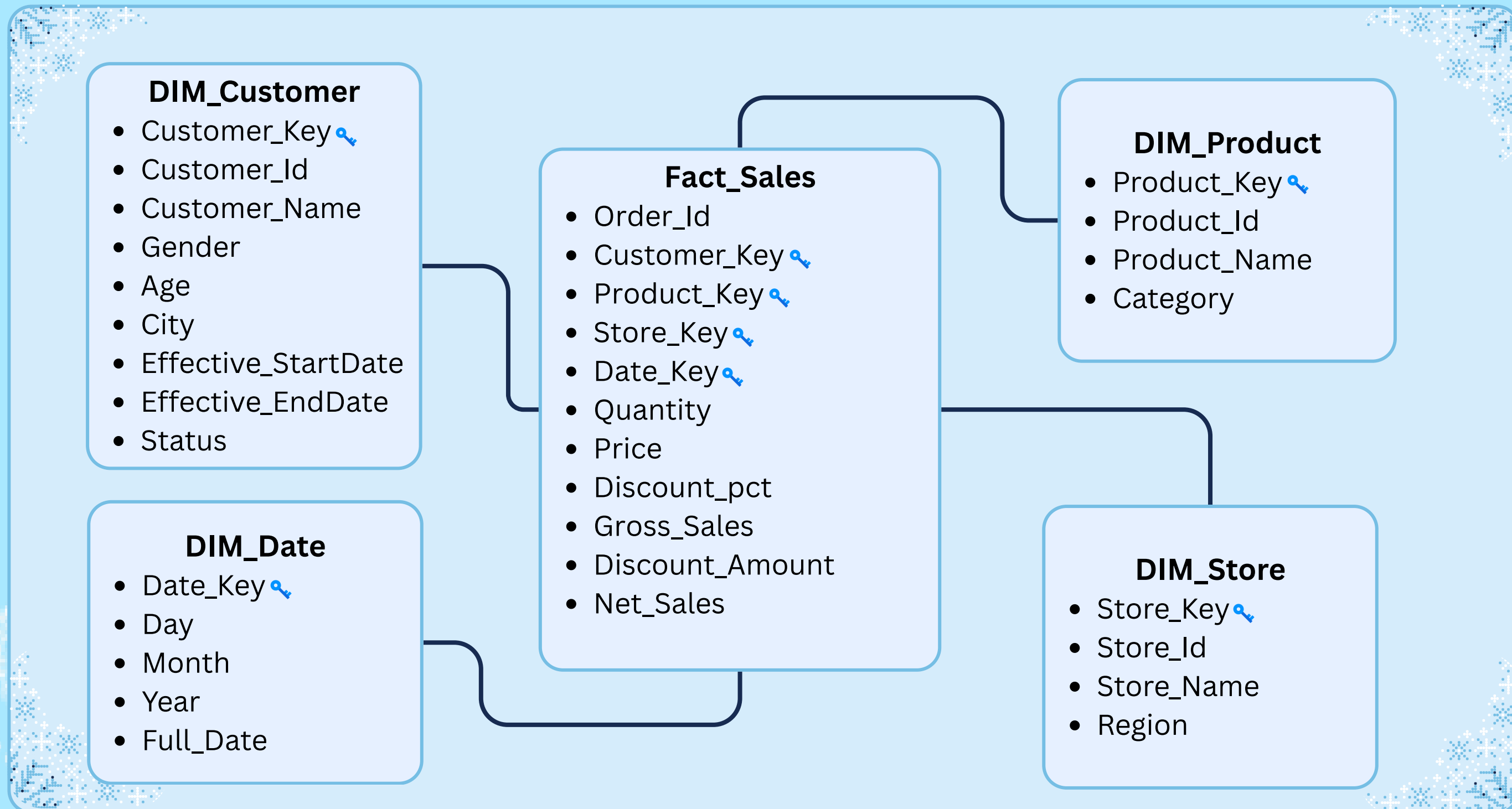


Solution - Architecture Diagram





Data Modeling





Dimension & Fact Table

Fact Table (FACT_SALES)

Purpose

- Stores quantitative business metrics generated from transactions

Why we use it

- Central table for analytics and KPIs
- Enables aggregation such as SUM, AVG, COUNT
- Supports revenue, volume, and discount analysis

Key Characteristics

- Contains foreign keys to dimensions
- Large and continuously growing
- One row per transaction

DIM Table (DIM_*)

Purpose

- Dimension tables store descriptive information that helps explain and analyze the data in the fact tables.

Dimension Tables in This Model

-  **DIM_Customer** - Stores customer details such as Name, Gender, Age, City
-  **DIM_Store** - Stores Store Name and Region details
-  **DIM_Product** - Stores product details such as product name, category.
-  **DIM_Date** - Stores date attributes like day, month, and year to support time-based analysis.





Data Loading (ETL Operation)

Extract

- Source data is received as CSV files
- Files are uploaded to Snowflake internal stage
- Data is loaded into Bronze tables using COPY INTO Command

Transform

- Raw data is cleaned and standardized
- Data types are validated and converted
- Invalid and null records are filtered
- Incremental changes are captured using Streams
- Transformations are automated using Tasks

Load

- Cleaned data is loaded into Silver tables
- Aggregated and business-ready data is created in Gold tables
- Gold layer is used for reporting and analysis





Reports & Dashboards

Report: Top Selling Product

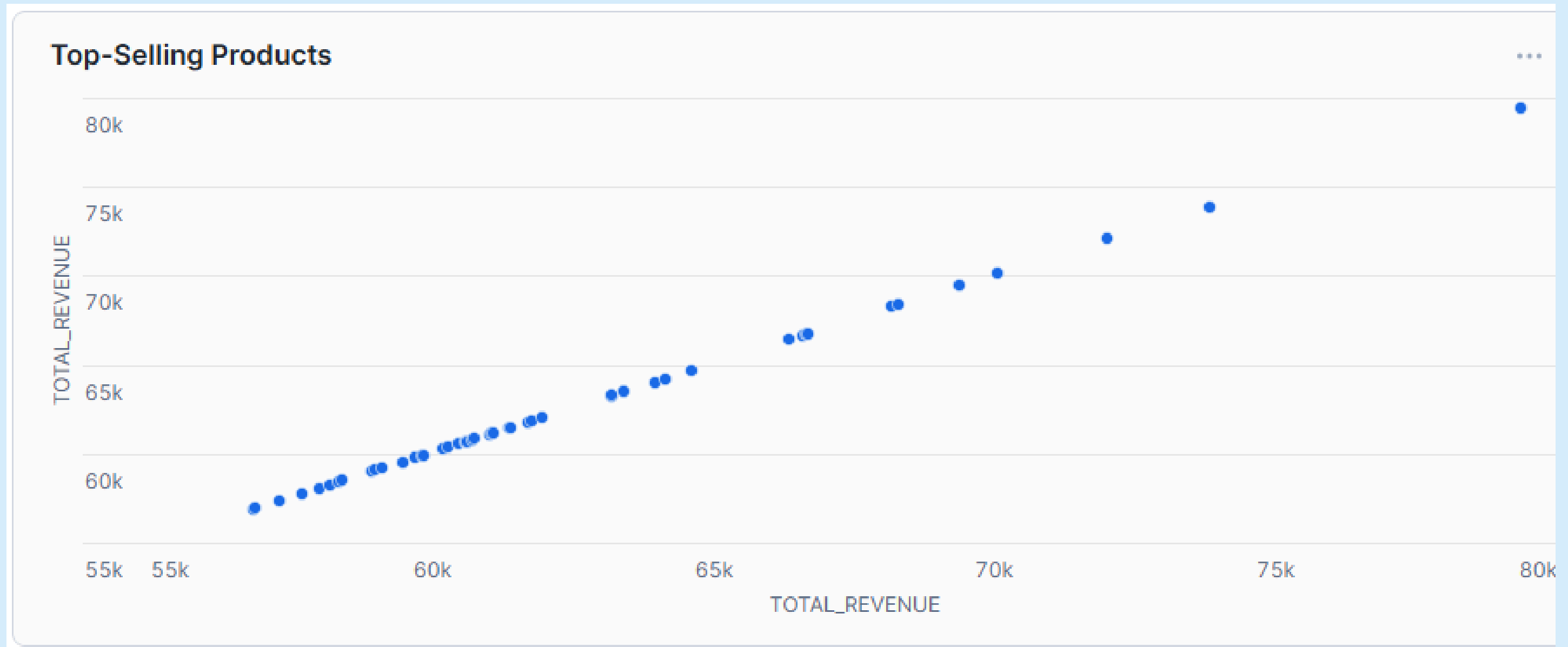
Top-Selling Products				50 rows	...
PRODUCT_NAME	CATEGORY	TOTAL_REVENUE	TOTAL_QUANTITY_SOLD		
Product_23205	Electronics	79381	19		
Product_14364	Clothing	73828	16		
Product_19344	Home	72015	16		
Product_9138	Grocery	70077	16		
Product_13530	Clothing	69392	19		
Product_14679	Electronics	68292	16		
Product_22292	Sports	68186	15		
Product_23968	Clothing	66704	15		
Show 42 more					





Reports & Dashboards

Dashboard: Top Selling Product





Reports & Dashboards

Report: Total Sales Revenue

Total_Sales_Revenue				100 rows	...
STORE_NAME	REGION	STORE_SALES_REVENUE	STORE_ORDER_VOLUME		
Store_28	East	2051271	283		
Store_27	North	1995032	254		
Store_3	West	1946097	271		
Store_58	South	1940503	286		
Store_91	East	1939424	271		
Store_56	East	1927167	273		
Store_98	East	1920199	257		
Store_19	North	1908280	292		
Show 92 more					





Reports & Dashboards

Dashboard: Total Sales Revenue

Total_Sales_Revenue		...
Store_1	1797183	
Store_10	1856286	
Store_100	1894027	
Store_11	1679744	
Store_12	1749767	
Store_13		1395236





*Thank
you!*

